Introduction

Bray-Curtis dissimilarity is the default index for comparing ecological communities described by species-abundance data. It ranges from 0 when two samples share the same composition and abundances to 1 when they share no species at all, and unlike binary indices it makes use of the abundance information rather than collapsing it to presence/absence. Microbial ecology, plant community work, and any compositional dataset where rare and abundant species both contribute meaningful information typically reach for Bray-Curtis as the first dissimilarity to compute.

Prerequisites

A working knowledge of distance and dissimilarity matrices, abundance data structures, and ordination methods (NMDS, PCoA) that consume dissimilarity matrices.

Theory

For two samples i and j with abundances x_{ik} across species k:

$$\mathrm{BC}_{ij} = \frac{\sum_k |x_{ik} - x_{jk}|}{\sum_k (x_{ik} + x_{jk})}.$$

It is a dissimilarity, not a strict metric — the triangle inequality can fail, so Bray-Curtis matrices are best visualised with non-metric MDS (NMDS) rather than principal coordinates. The binary form (presence-absence input) reduces to the Sørensen dissimilarity. Bray-Curtis is sensitive to dominant species; a log or Hellinger transformation of the abundance matrix before computing the index down-weights dominance and lets rare species contribute more.

Assumptions

Count or abundance data, all entries non-negative, comparable sampling effort across samples (or normalisation to relative abundance to remove the effort difference).

R Implementation

```r
library(vegan)

data(dune)
d_bc <- vegdist(dune, method = "bray")
as.matrix(d_bc)[1:3, 1:3]

# Log-transform to down-weight dominant species
d_bc_log <- vegdist(log1p(dune), method = "bray")

# NMDS visualisation
nmds <- metaMDS(dune, distance = "bray", k = 2, trymax = 50)
plot(nmds, type = "t")
```

Output & Results

`vegdist()` returns a `dist` object containing the lower-triangular Bray-Curtis matrix. `metaMDS()` runs NMDS on the resulting matrix and returns coordinates, stress, and convergence diagnostics. Stress values below 0.1 indicate excellent representation; 0.1–0.2 acceptable; above 0.2 poor.

Interpretation

A reporting sentence: "Bray-Curtis dissimilarity averaged 0.55 across all pairs (range 0.32 to 0.84), indicating substantial community turnover; NMDS in two dimensions converged at stress 0.07, and PERMANOVA (`vegan::adonis2`) detected a significant management-type effect ($F = 3.6$, $p = 0.001$, $R^2 = 0.21$)." Always pair Bray-Curtis ordination with a formal group test such as PERMANOVA or ANOSIM.

Practical Tips

- Log-transform abundance (`log1p()`) or use Hellinger distance (`vegdist(..., method = "hellinger")`) to reduce dominance by abundant species.
- Bray-Curtis is the default distance in `vegan::metaMDS()`; passing the data matrix directly is equivalent to `vegdist(data, "bray")` followed by `metaMDS()` on the dist.
- For phylogenetic-aware distances on microbial communities, use UniFrac (`GUniFrac::GUniFrac`) instead; it accounts for evolutionary relationships among taxa.
- Pair NMDS with PERMANOVA (`adonis2`) for group comparison and PERMDISP (`betadisper`) to check homogeneity of multivariate dispersion; significant PERMDISP can confound PERMANOVA conclusions.
- `vegdist()` supports many alternatives — Jaccard, Manhattan, Euclidean, Canberra, Gower, Hellinger — change `method` to switch.
- For very large community matrices, sparse implementations in `parallelDist` scale Bray-Curtis computation across cores without code changes.

R Packages Used

`vegan` for `vegdist()`, `metaMDS()`, `adonis2()`, and `betadisper()`; `GUniFrac` for phylogenetic UniFrac variants on microbiome data; `parallelDist` for fast multi-core distance computation; `phyloseq` for microbiome workflows that integrate Bray-Curtis with sequence and tree data.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE